

PHSX 765: Homework #1

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Problem 1

Unitary transformations: Position vs. Momentum basis.

Constructing unitary transformations between position and momentum spaces.

- (a) Consider a particle on a circle of length L . Define the particle's position eigenstates as $|x\rangle$ ($-L/2 < x \leq L/2$) and assume they form a complete orthonormal basis, i.e.,

$$\langle x|x'\rangle = \delta(x - x'),$$

$$\int_{-L/2}^{L/2} |x\rangle\langle x| dx = I.$$

Construct the complete orthonormal basis of discrete momentum eigenstates,

$$|p_k\rangle = \int_{-L/2}^{L/2} dx U(p_k, x) |x\rangle,$$

where

$$p_k = \frac{2\pi k\hbar}{L}, \quad k = 0, \pm 1, \pm 2, \dots,$$

such that

$$\langle p_k|p_{k'}\rangle = \delta_{k,k'}.$$

The "matrix" $U(p_k, x)$ can be viewed as the unitary transformation that implements the change of basis. Also find the inverse transformation $V(x, p_k)$ such that

$$|x\rangle = \sum_k V(x, p_k) |p_k\rangle.$$

Solution:

If any discrete basis $\{|n\rangle\}$ is orthonormal and complete, any state can be reached by:

$$|\psi\rangle = \sum_n c_n |n\rangle, \quad c_n = \langle n|\psi\rangle$$

- And I will note as a reminder to myself that orthonormality is $\langle n|n'\rangle = \delta_{nn'}$, and completeness is $\sum_n |n\rangle\langle n| = I$.
- As another note to myself, this is the same as projection $|\psi\rangle = \sum_n |n\rangle\langle n|\psi\rangle$.
- It's well known that $\langle x|\hat{p}|\psi\rangle = -i\hbar\frac{\partial}{\partial x}\langle x|\psi\rangle$

(Of course, for continuous states it's an integral instead of sum)

It follows that we want to express

$$U(p_k, x) = \langle x | p_k \rangle, \quad V(x, p_k) = \langle p_k | x \rangle$$

for $|p_k\rangle$ to be a momentum eigenstate, we must have

$$\begin{aligned} \hat{p} |p_k\rangle &= p_k |p_k\rangle \\ \langle x | \hat{p} |p_k\rangle &= p_k \langle x | p_k \rangle \\ \hat{p} \underbrace{\langle x | p_k \rangle}_{=U(p_k, x)} &= p_k \underbrace{\langle x | p_k \rangle}_{=U(p_k, x)} \\ -i\hbar \frac{\partial U(p_k, x)}{\partial x} &= p_k U(p_k, x) \\ U(p_k, x) &= \frac{1}{\sqrt{A}} e^{ip_k x / \hbar} \end{aligned}$$

Where, A is something like $\langle p_k | p'_k \rangle$

And the normalization constant is

$$\begin{aligned} \langle p_k | p'_k \rangle &= \int_{-\frac{L}{2}}^{\frac{L}{2}} \int_{-\frac{L}{2}}^{\frac{L}{2}} dx dx' U^*(p_k, x) U(p'_k, x) \\ &= \frac{1}{A} \int_{-\frac{L}{2}}^{\frac{L}{2}} \int_{-\frac{L}{2}}^{\frac{L}{2}} dx dx' e^{ip_k x / \hbar} e^{-ip'_k x / \hbar} \langle x | x' \rangle \\ &= \frac{1}{A} \int_{-\frac{L}{2}}^{\frac{L}{2}} dx e^{i(p_k - p'_k)x / \hbar} \end{aligned}$$

The normalization condition is $p_k - p'_k = 0$, otherwise the integral blows up. When it's zero, normalization is fulfilled by $A = L$.

The problem says when this is normalized, but I'd like to be explicit for the sake of review:

$$\begin{aligned} &= \frac{1}{A} \frac{e^{i(p_k - p'_k)x / \hbar}}{i(p_k - p'_k) / \hbar} \Bigg|_{x=-L/2}^{x=L/2} \\ &= \frac{1}{A} \frac{e^{i(p_k - p'_k)(L/2) / \hbar} - e^{i(p_k - p'_k)(-L/2) / \hbar}}{i(p_k - p'_k) / \hbar} \\ &= \frac{L}{A} \text{sinc} \left(\frac{(p_k - p'_k)L}{2\hbar} \right) \end{aligned}$$

And this vanishes when:

$$\frac{(p_k - p'_k)L}{2\hbar} = n\pi$$

so,

$$p_k - p'_k = \frac{2\pi\hbar n}{L}$$

In summary,

$$U(p_k, x) = \frac{1}{\sqrt{L}} e^{ip_k x / \hbar}, \quad V(p_k, x) = U(p_k, x)^* = \frac{1}{\sqrt{L}} e^{-ip_k x / \hbar}$$

- (b) Consider a particle on an infinite lattice whose positions are labeled by $x_n = na$. Assume the corresponding position eigenkets $|n\rangle$ form a complete orthonormal basis, i.e.,

$$\langle n|n'\rangle = \delta_{n,n'},$$

$$\sum_n |n\rangle\langle n| = I.$$

Construct the complete orthonormal basis of momentum eigenstates,

$$|p\rangle = \sum_n U(p, n)|n\rangle,$$

where

$$-\frac{\pi\hbar}{a} < p \leq \frac{\pi\hbar}{a}$$

such that

$$\langle p|p'\rangle = \delta(p - p').$$

The "matrix" $U(p, n)$ can be viewed as the unitary transformation that implements the change of basis. Also find the inverse transformation $V(n, p)$ such that

$$|n\rangle = \int_{-\pi\hbar/a}^{\pi\hbar/a} dp V(n, p)|p\rangle.$$

Solution:

This is slightly different than (a) in the sense that it's periodic in a zone and not over a length L . We'd get a similar U , now just replacing $x \rightarrow na$.

$$U(p, n) = \frac{1}{\sqrt{A}} e^{ipna/\hbar}$$

The other important distinction is that n is discrete so $|p\rangle$ must be written as a sum normalized when,

$$\begin{aligned} A &= \langle p|p'\rangle \\ &= \sum_{n=-\infty}^{\infty} \sum_{n'=-\infty}^{\infty} e^{ipna/\hbar} e^{-ip'na/\hbar} \langle n|n'\rangle \\ &= \sum_{n=-\infty}^{\infty} e^{i(p-p')na/\hbar} \end{aligned}$$

I think that switching $V = U^\dagger$ to get an integral and evaluate $|n\rangle$ instead of $|k\rangle$ is not valid, since that's doing unknown things to the bounds if we weren't told this. Some googling tells me this is the Poisson sum which is obviously well known since there is such a thing as the discrete Fourier transform. This also interestingly nets the Dirac comb.

$$\sum_{n=-\infty}^{\infty} e^{iqn} = 2\pi \sum_{m=-\infty}^{\infty} \delta(q - 2\pi m), \quad q \equiv \frac{(p' - p)a}{\hbar}$$

Because

$$-\frac{\pi\hbar}{a} < p \leq \frac{\pi\hbar}{a}$$

only $m = 0$ need be considered. Thus, it's normalized by

$$A = \frac{2\pi a}{\hbar}$$

Practically speaking we can jump between conjugate spaces by a fourier transform, but more rigorously,

$$\langle n|p\rangle = \langle p|n\rangle^* \implies \boxed{V(n, p) = \sqrt{\frac{a}{2\pi\hbar}} e^{-ipna/\hbar}}$$

Problem 2

Particle on a circular lattice

Let $|\phi_n\rangle$, $n = 0, 1, 2, \dots, N-1$ define the position eigenstates of a particle on a circle where

$$\phi_n = \frac{2\pi n}{N}$$

label the angles the particle can be located at. Assume these states form a complete orthonormal basis.

(a) Define the lattice translation operator

$$T|\phi_n\rangle = |\phi_{n+1}\rangle, \quad T|\phi_{N-1}\rangle = |\phi_0\rangle,$$

which is the natural definition if the wave-functions are periodic. Show that

$$|k\rangle = \frac{1}{\sqrt{N}} \sum_n e^{-i\phi_n k} |\phi_n\rangle$$

are eigenkets of T . Show that the corresponding eigenvalues can be expressed as $\exp(-ip_k)$ where p_k can be defined as the lattice momentum. Find the allowed values of p_k .

Solution:

This is the same as being asked to prove that the translation operator is still the infinitesimal generator of the momentum operator on the lattice.

$$\begin{aligned} T|k\rangle &= \frac{1}{\sqrt{N}} \sum_n^{N-1} e^{-i\phi_n k} T|\phi_n\rangle \\ &= \frac{1}{\sqrt{N}} \sum_n^{N-1} e^{-i\phi_n k} |\phi_{n+1}\rangle \\ &= \frac{1}{\sqrt{N}} \sum_{n'=0}^{N-1} e^{-i2\pi(n'-1)k/N} |\phi_{n'}\rangle \quad (\text{Let } n' = n + 1) \\ &= \left(e^{i2\pi k/N}\right) \left(\frac{1}{\sqrt{N}}\right) \sum_{n'=0}^{N-1} e^{-i2\pi n'k/N} |\phi_{n'}\rangle \\ &= e^{i2\pi k/N} |k\rangle \end{aligned}$$

So, $|k\rangle$ is an eigenket of T with eigenvalue $e^{2\pi i k/N} = e^{-ip_k}$ if

$$p_k = -\frac{2\pi k}{N}$$

To continue and find the allowed k , it's good to remember that the normalization condition $\langle k|k'\rangle =$ should be something finite, which is a constraint. Let n' now represent a dummy index:

$$\begin{aligned} S &= \sum_n \sum_n' e^{-i\phi_n k} e^{i\phi_{n'} k} \langle \phi_n | \phi_{n'} \rangle \\ &= \sum_n e^{-i(\phi_n - \phi_{n'})k} \\ &= \sum_n e^{-2\pi i(n-n')k/N} \end{aligned}$$

So values ought to be integers, but it's also obvious that there's a wraparound, given the circular nature,

$$e^{-2i\pi n(k+N)/N} = e^{-2i\pi nk/N} e^{-2i\pi n} = e^{-2i\pi nk}$$

So,

$$p_k = \exp\left(\frac{i2\pi k}{N}\right); \quad k = 0, 1, 2, \dots, N-1$$

- (b) Show that while T is not Hermitian, it is unitary. What are the eigenvalues and eigenkets of $T^\dagger$, which can be viewed as a backward translation operator?

Solution:

A unitary operator is one that satisfies:

$$UU^\dagger = U^\dagger U = I$$

Or equivalently

$$UU^{-1} = I$$

We can expect that $T^\dagger$ will act in the other direction, given the nature of the translation operator:

$$T^\dagger |\phi_n\rangle = |\phi_{n-1}\rangle$$

Then,

$$\begin{aligned} \langle \phi_n | TT^\dagger | \phi'_n \rangle &\stackrel{?}{=} \langle \phi_n | T^\dagger T | \phi'_n \rangle \\ \langle \phi_n | T | \phi'_{n-1} \rangle &\stackrel{?}{=} \langle \phi_n | T^\dagger | \phi'_{n+1} \rangle \\ \langle \phi_n | \phi'_n \rangle &= \langle \phi_n | \phi'_n \rangle \end{aligned}$$

And because $TT^\dagger |\phi_n\rangle = |\phi_n\rangle$, and $|\phi_n\rangle$ diagonalizes a matrix, the translation operator is unitary and not just normal. Since it's unitary it will have the same eigenkets and eigenvalues with conjugate sign:

$$p_k = \exp\left(-\frac{i2\pi k}{N}\right)$$

One could also just say both chains return to the original state.

(c) Redefine the translation operator as

$$T|\phi_n\rangle = |\phi_{n+1}\rangle, \quad T|\phi_{N-1}\rangle = -|\phi_0\rangle,$$

which is the natural definition with anti-periodic boundary conditions. Find the eigenstates and eigenvalues of T in this case. Find the allowed lattice momentum p_k .

Solution:

I would imagine the correct way to go about this is to consider what happens to the summation limits first. For all n except $N - 1$, we could get the same answer as (a). This necessitates writing a sum plus the last term. If I'm thinking in terms of the matrix, it's like the last diagonal term has a flipped sign.

$$\begin{aligned} T|k\rangle &= \frac{1}{\sqrt{N}} \sum_n^{N-1} e^{-i\phi_n k} T|\phi_n\rangle \\ \sqrt{N}T|k\rangle &= \left[\sum_n^{N-2} e^{-i\phi_n k} T|\phi_n\rangle \right] + e^{-i\phi_{N-1} k} T|\phi_{N-1}\rangle \\ &= \left[\sum_n^{N-2} e^{-i\phi_n k} |\phi_{n+1}\rangle \right] - e^{-i\phi_{N-1} k} |\phi_0\rangle \end{aligned}$$

In the case $n = 0$

$$\begin{aligned} &= e^{-i\phi_0 k} |\phi_1\rangle - e^{-i\phi_{N-1} k} |\phi_0\rangle \\ &= \cancel{e^{-i(2\pi(0))k/N} |\phi_1\rangle} - e^{-i\phi_{N-1} k} |\phi_0\rangle \\ &= -e^{-i2\pi(N-1)k/N} |\phi_0\rangle \end{aligned}$$

We know this must be the same as the other end,

$$\begin{aligned} e^{2i\pi k/N} &= -e^{-i2\pi k} e^{i2\pi k/N} && \text{(First and last terms)} \\ \implies -1 &= e^{-2i\pi k} \end{aligned}$$

k that satisfy this are $k = m + \frac{1}{2}$ for $m = 0, 1, 2, 3, \dots$

Thus, p_k are:

$$\boxed{-\frac{2\pi k}{N}, \quad k = m + \frac{1}{2}}$$

Problem 3

Free Particle on a Circular Lattice

We learned that the Hamilton operator of a free particle on a circular lattice with periodic boundary conditions can be described through the Hamilton operator

$$H_{n',n} = \epsilon (2\delta_{n',n} - \delta_{n'+1,n} - \delta_{n'-1,n}),$$

where

$$\epsilon = \frac{N^2 \hbar^2}{8\pi^2 m R^2}$$

and $n, n' = 0, 1, 2, \dots, N-1$ such that $n, n' = N$ is the same site as $n, n' = 0$.

- (a) Compute numerically (using Mathematica or another convenient tool) the eigenvalues and eigenvectors of H for $N = 10$ (i.e., a 10×10 matrix). Note that the eigenvalues are obtained from solving the eigenvalue equation:

$$\sum_n H_{n',n} \psi_n^{(k)} = \epsilon_k^L \psi_{n'}^{(k)}.$$

Verify these are correct after solving part (b) below.

Solution:

$$\epsilon_k^L = \epsilon [0.0000 \quad 0.3820 \quad 0.3820 \quad 1.3820 \quad 1.3820 \quad 2.6180 \quad 2.6180 \quad 3.6180 \quad 3.6180 \quad 4.0000]^T$$

$$|\psi_n^{(k)}\rangle = \epsilon \begin{bmatrix} 0.3162 & -0.4423 & 0.0663 & -0.4472 & 0.0000 & 0.4472 & 0.0053 & 0.0019 & 0.4472 & 0.3162 \\ 0.3162 & -0.3968 & -0.2064 & -0.1382 & 0.4253 & -0.1331 & -0.4269 & 0.2614 & -0.3629 & -0.3162 \\ 0.3162 & -0.1997 & -0.4002 & 0.3618 & 0.2629 & -0.3649 & 0.2586 & -0.4247 & 0.1400 & 0.3162 \\ 0.3162 & 0.0736 & -0.4411 & 0.3618 & -0.2628 & 0.3587 & 0.2671 & 0.4259 & 0.1364 & -0.3162 \\ 0.3162 & 0.3189 & -0.3136 & -0.1382 & -0.4253 & 0.1432 & -0.4237 & -0.2644 & -0.3607 & 0.3162 \\ 0.3162 & 0.4423 & -0.0663 & -0.4472 & 0.0000 & -0.4472 & -0.0053 & 0.0019 & 0.4472 & -0.3162 \\ 0.3162 & 0.3968 & 0.2064 & -0.1382 & 0.4253 & 0.1331 & 0.4269 & 0.2614 & -0.3629 & 0.3162 \\ 0.3162 & 0.1997 & 0.4002 & 0.3618 & 0.2629 & 0.3649 & -0.2586 & -0.4247 & 0.1400 & -0.3162 \\ 0.3162 & -0.0736 & 0.4411 & 0.3618 & -0.2628 & -0.3587 & -0.2671 & 0.4259 & 0.1364 & 0.3162 \\ 0.3162 & -0.3189 & 0.3136 & -0.1382 & -0.4253 & -0.1432 & 0.4237 & -0.2644 & -0.3607 & -0.3162 \end{bmatrix}$$

(b) By noting that

$$H \equiv \epsilon (2 - T^\dagger - T),$$

where T is the translation operator defined earlier, solve for the eigenvalues ϵ_k^L and eigenfunctions $\psi_n^{(k)}$ so that the eigenkets are defined by

$$|\epsilon_k^L\rangle = \sum_n \psi_n^{(k)} |\phi_n\rangle$$

for an arbitrary N .

Solution:

$$\begin{aligned} H |\epsilon_k^L\rangle &= \epsilon \sum_n (2 - T^\dagger - T) \psi_n^{(k)} |\phi_n\rangle \\ &= \epsilon \sum_n \left(2\psi_n^{(k)} |\phi_n\rangle - (T^\dagger \psi_n^{(k)} |\phi_n\rangle) - (T \psi_n^{(k)} |\phi_n\rangle) \right) \\ &= \epsilon \sum_n \left(2\psi_n^{(k)} - \psi_{n+1}^{(k)} - \psi_{n-1}^{(k)} \right) |\phi_n\rangle \end{aligned}$$

Before I justified the form of the eigenket for a particle on a circle,

$$\psi_n^k = e^{-2\pi i n k / N}$$

so,

$$\begin{aligned} &= 2\psi_n^{(k)} - e^{-i2\pi k(n+1)/N} - e^{-i2\pi k(n-1)/N} \\ &= 2\psi_n^{(k)} - e^{-i2\pi k/N} e^{-i2\pi k n / N} - e^{i2\pi k/N} e^{-i2\pi k n / N} \\ &= 2\psi_n^{(k)} - e^{-i2\pi k/N} \psi_n^{(k)} - e^{i2\pi k/N} \psi_n^{(k)} \\ &= \psi_n^{(k)} \left(2 - e^{-i2\pi k/N} - e^{i2\pi k/N} \right) \end{aligned}$$

The eigenvalue problem becomes:

$$\begin{aligned} H |\epsilon_k^L\rangle &= \epsilon \sum_n \left(2 - e^{-i2\pi k/N} - e^{i2\pi k/N} \right) \psi_n^{(k)} |\phi_n\rangle \\ H |\epsilon_k^L\rangle &= \epsilon \left(2 - e^{-i2\pi k/N} - e^{i2\pi k/N} \right) \sum_n \psi_n^{(k)} |\phi_n\rangle \\ H |\epsilon_k^L\rangle &= \epsilon \left(2 - e^{-i2\pi k/N} - e^{i2\pi k/N} \right) |\epsilon_k^L\rangle \end{aligned}$$

So the energy eigenvalues are the constant term

$$\begin{aligned} \epsilon_k^L &= \epsilon (2 - e^{ip_x} - e^{-ip_x}) \\ &= \epsilon (2 - 2 \cos p_k) \end{aligned}$$

and has computed values:

$$\epsilon_k^L = \epsilon [0.0 \quad 0.3820 \quad 0.3820 \quad 1.3820 \quad 1.3820 \quad 2.6180 \quad 2.6180 \quad 3.6180 \quad 3.6180 \quad 4.0]^T$$

Problem 4

Particle in a Potential on a Circle

Consider the Hamilton operator of a particle on a circle given by

$$H = \frac{L_{\phi_1}^2}{2mR^2} + m\omega^2 R^2(1 - \cos \phi).$$

Using lattice regularization, compute numerically the lowest three eigenvalues (up to 1% accuracy) when

$$\lambda = \frac{m\omega R^2}{\hbar} = 1, 10, 50.$$

What do you expect your answers will converge to in the limit $\lambda \rightarrow \infty$?

Solution:

Rightmost term can be parametrized as $\phi \rightarrow n\Delta\phi$, remembering that $\Delta\phi \equiv \frac{2\pi}{N}$ and $n = 0, 1, 2, \dots, N-1$. I'll label the left and right of the sum $\hat{U}$ and $\hat{V}$, respectively

$$H = \underbrace{\frac{L_{\phi_1}^2}{2mR^2}}_{\hat{U}} + \underbrace{m\omega^2 R^2(1 - \cos \phi)}_{\hat{V}}$$

Easy enough to write as a matrix:

$$\begin{aligned} \langle \phi_n | \hat{V} | \phi'_n \rangle &= m\omega^2 R^2(1 - \cos \phi) \delta(\phi - \phi') \\ &= \hbar\lambda \left(1 - \cos \left[\frac{2\pi n}{N} \right] \right) \delta(\phi - \phi') \end{aligned}$$

Result:

$$\begin{aligned} \lambda = 1 &\implies [0.46491035 \quad 1.34325956 \quad 1.85353102] \\ \lambda = 10 &\implies [0.49655291 \quad 1.48270779 \quad 2.45486992] \\ \lambda = 50 &\implies [0.49783372 \quad 1.48917185 \quad 2.47185580] \end{aligned}$$

Appears to converge to $\frac{n}{2}$, where $n = 1, 3, 5, \dots$

Problem 5

Two interacting particles on a circle

Consider the Hamilton operator for two distinguishable particles on a circle given by

$$H = \frac{L_{\phi_1}^2}{2mR^2} + \frac{L_{\phi_2}^2}{mR^2} + m\omega^2 R^2 ((1 - \cos \phi_1) + (1 - \cos \phi_2) + (1 - \cos(\phi_1 - \phi_2))).$$

Using lattice regularization, compute numerically the ground state (up to 1% accuracy) when

$$\frac{m\omega R^2}{\hbar} = 1.$$

Solution:

$$\hat{U}_{\phi_1} + 2\hat{U}_{\phi_2} + \hat{V}_{\phi_1} + \hat{V}_{\phi_2} + \text{cross term}$$

This cross term is diagonal (imagine evaluating it in a bracket, the outer products would be delta functions) with terms like

$$1 - \cos(\phi_1 - \phi_2) \delta_{nn'} \delta_{mm'}$$

The decoupled terms are quite easy to generate, since I can simply do:

$$\hat{U}_{\phi_1} \otimes I + I \otimes 2\hat{U}_{\phi_2} + \hat{V}_{\phi_1} \otimes I + I \otimes \hat{V}_{\phi_2}$$

Where $\otimes$ is just `np.kron()` and I is `np.eye(N)`. Some of the unique values are:

$$\epsilon_k^L = [1.4812 \quad 4.9763 \quad 6.8505 \quad 8.2675 \quad 9.554 \quad 10.7886 \quad 12.0882 \quad 13.6508 \quad 14.5719 \quad 16.4355]^T$$

Seems like the ground state is 1.4812 (my poor laptop takes 3 minutes to do this calculation with $N=100$).

Note, I can't really run a more precise computation on my laptop in a reasonable time. If I were to check the accuracy I could perhaps write an iterative solver or compare the delta between matrices of size N and $N-1$.

Code for 3a:

```
1  import numpy as np
2
3  s = 10
4
5  # generate some off-axis diagonal ones
6  lower = np.zeros((s,s),dtype=np.float64)
7  upper = np.zeros((s,s),dtype=np.float64)
8  np.fill_diagonal(lower, -1, wrap=False)
9  np.fill_diagonal(upper, -1, wrap=False)
10 lower = np.roll(lower, -1, axis=1)
11 upper = np.roll(upper, 1, axis=1)
12
13 H = np.zeros((s,s), dtype=np.float64)
14 H = lower + upper
15
16 # on-axis ones
17 np.fill_diagonal(H, 2, wrap=False)
18
19 print(np.linalg.eigh(H).eigenvalues)
20 print()
21 print(np.linalg.eigh(H).eigenvectors)
22
```

Code for 3b:

```
1  import numpy as np
2
3  N = 10
4  k = np.arange(0, 10)
5  pk = (2 * np.pi * k) / N
6
7  a_res = np.sort(2 * (1 - np.cos(pk)))
8
9  print(a_res)
10
```

Code for 4:

```
1  import numpy as np
2
3  N = 200
4  def hamiltonian(lam):
5      eps = N**2 / (8 * np.pi**2 * lam)
6
7      lower = np.zeros((N, N), dtype=np.float64)
8      upper = np.zeros((N, N), dtype=np.float64)
9      np.fill_diagonal(lower, -1)
10     np.fill_diagonal(upper, -1)
11     lower = np.roll(lower, -1, axis=1)
12     upper = np.roll(upper, 1, axis=1)
13     U = lower + upper
14     np.fill_diagonal(U, 2)
15     U *= eps
16
17     n = np.arange(N)
18     phi_n = 2 * np.pi * n / N
19     V = lam * np.diag(1 - np.cos(phi_n))
20
21     return U + V
22
23 for lam in [1, 10, 50]:
24     H = hamiltonian(lam)
25     eigvals = np.linalg.eigvalsh(H)
26     print(f"eigvals = {np.round(eigvals[:3], 4)}")
27
```

Code for 5:

```

1  import numpy as np
2
3  N = 100
4
5  lam = 1
6  eps = N**2 / (8 * np.pi**2 * lam)
7
8  lower = np.zeros((N, N), dtype=np.float64)
9  upper = np.zeros((N, N), dtype=np.float64)
10 np.fill_diagonal(lower, -1)
11 np.fill_diagonal(upper, -1)
12 lower = np.roll(lower, -1, axis=1)
13 upper = np.roll(upper, 1, axis=1)
14
15 U = lower + upper
16 np.fill_diagonal(U, 2)
17 U *= eps
18
19 n = np.arange(N)
20 phi_n = 2 * np.pi * n / N
21 V = lam * np.diag(1 - np.cos(phi_n))
22
23 # ---
24
25 I = np.eye(N)
26
27 U1 = np.kron(U, I)
28 U2 = np.kron(I, 2*U)
29 V1 = np.kron(V, I)
30 V2 = np.kron(I, V)
31
32 # cross terms are generated by "brute force" instead of an otimes
33 phi1_grid, phi2_grid = np.meshgrid(phi_n, phi_n, indexing='ij')
34 V12 = lam * (1 - np.cos(phi1_grid - phi2_grid))
35 V12 = np.diag(V12.flatten()) # fix the shape
36
37 # ---
38
39 H = (U1 + U2) + (V1 + V2) + V12
40 eigvals = np.linalg.eigvalsh(H)
41 print(f"eigvals = {np.round(eigvals[0:N], 4)}") # N degeneracy?!
42

```